

Complex Python

Lesson #3

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Functions

Functions take in inputs and give an output based on the code.

```
def hello(name):  2+ usages
    return f"Hello, {name}!"

print(hello("John"))
print(hello("Jane"))
```

```
Hello, John!
Hello, Jane!
```

def is short for define, it lets Pycharm know that this shouldn’t be ran now but only when the function is called upon. name is the parameter or input. return sends the code to the program to wun when needed.

Now try calculating your own function!

Lists

Lists are an ordered collection of items that can be changed. When you look at a value in the list, the 1st value will be indexed as 0, then 1, 2, etc.

```
fruits = ["apple", "banana", "cherry"]
print(fruits[1])
```

```
banana
```

Changing the item can be done by doing list[0] = value

```
fruits = ["apple", "banana", "cherry"]
print(fruits[1])
fruits[1] = "orange"
print(fruits[1])
```

```
banana
orange
```

Tuples

```
fruits = ("apple", "banana", "cherry")
print(fruits[1])
```

```
banana
```

Tuples contain values that do not change at all. In every other aspect they are similar to lists. The syntax for this instead of brackets is parentheses. Tuples are more efficient and and ran faster by the computer than any other thing, leading it to be used as an output of functions.

Try it yourself!

Dictionaries

Dictionaries are useful as you can attach a label to every value. You can add more values to the dictionary.

```
def get_user():  1 usage
    return {"name": "Jane", "age": 28}

user = get_user()
print(user["name"])
print(user["age"])
```

```
Jane
28
```

```
database1 = {"John" : ({"Tall", "Nice"}, 15)}
traits, age = database1["John"]
print(traits)
```

```
{'Tall', 'Nice'}
```

Sets

set are a very quick and efficient way to check if there are duplicates in a data set. It can be used to make sure a user doesn’t fllow someone multiple times of social media.

```
def getUniqueColors():  1 usage
    return {"red", "blue", "red", "green"}


print(getUniqueColors())
```

```
{'blue', 'green', 'red'}
```

Activity Time!

Create code that has info about a student that uses at least 1 dictionary, 1 Tuple, and either 1 List or 1 Set. Also add a function that can return this info, when the students name is inputted.

My Code

```
database = {"John" : ({"Algebra", "Apes"}, (11,2,2010)), "Alice" : ({"Calculus", "Biology"}, (7,6,2010))}

def getStudent(name):  1 usage
    if name in database:
        classes, dateOfBirth = database[name]
        print(classes)
        return print(dateOfBirth)
    else:
        return print("Error, user not found")

getStudent("Alice")
```

```
{'Biology', 'Calculus'}
(7, 6, 2010)
```

Thank You!

Any Questions?